Combining terms of like powers of h, we have

$$P = p + hF + \frac{h^2}{2}pF' + \frac{h^3}{4}\left[F'F + p^2F''\right] + \mathcal{O}(h^4).$$

We carry these terms to compare against the expansion of the exact solution. Since $\dot{q} = p$, we have $\ddot{q} = \dot{p} = F(q)$, and the third derivative is $q^{(3)} = F'(q)p$. On the other hand $\dot{p} = F(q)$ implies that $\ddot{p} = F'(q)\dot{q} = pF'$, and thus

$$p^{(3)} = p^2F'' + F'F.$$

The Taylor expansion of the solution is (taking $q(t) = q, p(t) = p$):

$$q(t+h) = q + hp + \frac{h^2}{2}F + \frac{h^3}{6}F'p + \mathcal{O}(h^4),$$
$$p(t+h) = p + hF + \frac{h^2}{2}pF' + \frac{h^3}{6}\left[p^2F'' + F'F\right] + \mathcal{O}(h^4).$$

We now examine the series expansions for the exact and Verlet solutions and find that these differ in the third (and higher) order terms.

$$Q - q(t+h) = \frac{h^3}{6}F'p + \mathcal{O}(h^4),$$

and

$$P - p(t+h) = \frac{h^3}{12}\left[p^2F'' + F'F\right] + \mathcal{O}(h^4).$$

These relations can be summarized as telling us that

$$\|\mathscr{G}_h(z) - \mathscr{F}_h(z)\| = \kappa(z)h^3 + \mathcal{O}(h^4),$$

where $\kappa(z) = \kappa(q,p)$ is a function of the position and momentum.

We may then define

$$\bar{K} = \max_{t\in[0,\tau]} \kappa(z(t))$$

bounding the local error by (with neglect of the fourth order terms) $\bar{K}h^3$. Thus the Verlet method is consistent of order two.

To complete the convergence proof for Verlet's method, we would still need to verify the second assumption. This requires the assumption that the force field $\boldsymbol{F}$ satisfy a Lipschitz condition:

$$\|\boldsymbol{F}(\boldsymbol{u}) - \boldsymbol{F}(\boldsymbol{w})\| \leq \hat{L}\|\boldsymbol{u} - \boldsymbol{w}\| \tag{2.15}$$

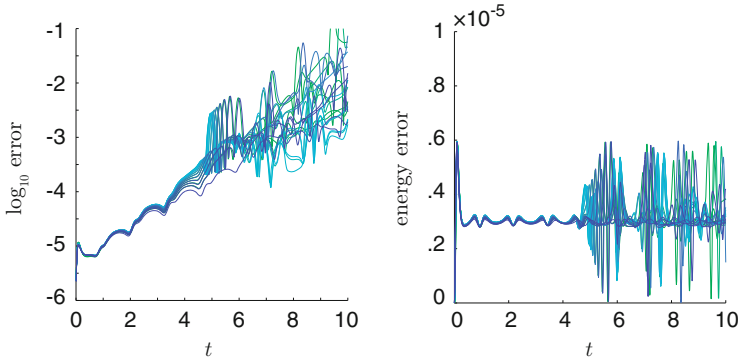


Fig. 2.4 The performance of the Verlet method is demonstrated by repeating the tests of error growth discussed for Euler's method (compare Fig. 2.3). *Left panel*: In the case of the Verlet method, the error is much smaller in an absolute sense than that for the Euler method with the same step-size, but the growth still appears to be exponential. *Right*: the energy error growth as a function of time, very remarkably, compared to the equivalent figure for Euler's method the energy error appears not to grow at all beyond some reasonably small limiting value, suggesting that the Verlet numerical approximation will remain near the desired energy surface

for all $\boldsymbol{u}, \boldsymbol{w}$. Generally speaking this could be taken to hold in a neighborhood of the solution where all approximate solutions for $h < \bar{h}$ are assumed to lie. With a bit of effort, it is then possible to demonstrate the stability condition for the numerical method (see Exercise 4).

In the left-hand panel of Fig. 2.4 we report the Verlet error growth as a function of time for the Lennard-Jones trimer. Verlet is substantially more accurate in this simulation, although the error still appears to grow exponentially.

Due to the chaotic nature of molecular dynamics, which implies a sensitivity to perturbations of the initial condition or the differential equations themselves, it is to be expected that the global error due to using a numerical method will always grow rapidly (exponentially) in time. As we shall see in later chapters, this does not necessarily mean that a long trajectory is entirely without value. In molecular dynamics it turns out that the real importance of the trajectory is that it provides a mechanism for calculating averages that maintain physical parameters. The simplest example of such a parameter is the energy.

What is more relevant for using Verlet to simulate molecular dynamics is the remarkable stability of energy shown in the right-hand panel of Fig. 2.4. Notice that the energy error does not appear to exhibit a steady accumulation in time (unlike for Euler's method, where it exhibited a linear-in-time growth). The explanation for this unexpected behavior lies in the structural properties of the method, a topic we explore in this and the next chapter.

As another illustration of the performance of the Verlet method, we mention that the dynamical trajectories given in the previous chapter (in particular those given Examples 1.8 and 1.9) were computed using this method (with stepsize $h = 0.00001$). In the case of the calculation of the exponential separation of